



DSA Marathon Master Roadmap

Organized by NextGen Data Minds Community

Community Milestone Program

Welcome to the NextGen Data Minds DSA Marathon! This custom schedule charts 160 days of intensive skill-building, pairing rigorous regular theory with tactical execution metrics to ensure supreme problem-solving mastery.

The Weekly Plan of Attack

Each weekly module enforces a comprehensive 6-day learning architecture designed to ensure deep retention and analytical precision:

Day	Tactical Strategy & Deliverables
Monday	Detailed Notes: Formulate extensive text breakdowns, logic patterns, and algorithmic blueprints.
Tuesday	Cheat Sheet: Map high-level summaries, syntax structures, space/time trade-offs, and critical templates.
Wednesday	Easy Problems: Complete 3-5 basic structural programming challenges to cultivate code execution speed.
Thursday	Advanced Concepts + Medium Problems: Navigate optimization pathways and tackle algorithmic patterns.
Friday	Practice Set: Focus on real-world company sets and complex problem variations.
Saturday	Weekly Quiz & Contest: Live timed testing sessions to measure accurate code production and retrieval speed.
Sunday	Review & Buffer Day: Debug failed test cases, balance missed timelines, and rest.

Chronological Roadmap & Milestones

Focus Area / Topic	Weekly Implementation Cycle
Basics & Fundamentals	Full 6-Day Learning & Problem-Solving Cycle
Conditional Statements	Full 6-Day Learning & Problem-Solving Cycle
Loops & Iteration	Full 6-Day Learning & Problem-Solving Cycle
Pattern Printing	Full 6-Day Learning & Problem-Solving Cycle
Functions & Scope	Full 6-Day Learning & Problem-Solving Cycle
Pointers & Memory Management	Full 6-Day Learning & Problem-Solving Cycle
Arrays (1D Arrays)	Full 6-Day Learning & Problem-Solving Cycle
Multidimensional Arrays	Full 6-Day Learning & Problem-Solving Cycle
Strings & Character Arrays	Full 6-Day Learning & Problem-Solving Cycle
Time & Space Complexity Analysis	Full 6-Day Learning & Problem-Solving Cycle
Recursion Basics	Full 6-Day Learning & Problem-Solving Cycle
Advanced Recursion Problems	Full 6-Day Learning & Problem-Solving Cycle
Sorting Algorithms I	Full 6-Day Learning & Problem-Solving Cycle
Sorting Algorithms II	Full 6-Day Learning & Problem-Solving Cycle
Prefix & Suffix Sum Techniques	Full 6-Day Learning & Problem-Solving Cycle
Sliding Window & Two Pointers	Full 6-Day Learning & Problem-Solving Cycle
Bitwise Operations	Full 6-Day Learning & Problem-Solving Cycle
Basic Maths	Full 6-Day Learning & Problem-Solving Cycle
Mathematics & Number Theory	Full 6-Day Learning & Problem-Solving Cycle
Object-Oriented Programming (OOPs)	Full 6-Day Learning & Problem-Solving Cycle
Linked Lists - Part 1	Full 6-Day Learning & Problem-Solving Cycle
Linked Lists - Part 2	Full 6-Day Learning & Problem-Solving Cycle
Stacks & Queues	Full 6-Day Learning & Problem-Solving Cycle
Trees & Binary Trees	Full 6-Day Learning & Problem-Solving Cycle

Focus Area / Topic	Weekly Implementation Cycle
Sets & Disjoint Set Union	Full 6-Day Learning & Problem-Solving Cycle
Hash Maps & Hashing	Full 6-Day Learning & Problem-Solving Cycle
Heaps & Priority Queues	Full 6-Day Learning & Problem-Solving Cycle
Dynamic Programming (DP)	Full 6-Day Learning & Problem-Solving Cycle
Greedy Algorithms	Full 6-Day Learning & Problem-Solving Cycle
Backtracking Algorithms	Full 6-Day Learning & Problem-Solving Cycle
Advanced Bit Manipulation	Full 6-Day Learning & Problem-Solving Cycle
Graph Algorithms	Full 6-Day Learning & Problem-Solving Cycle